

THE STARTING SITUATION

The transports are real. The planning is static.

Every day we move thousands of shipments via road, air and sea. Yet the planning of lead times and traffic relations is still often based on static assumptions.

Fixed lead times and kilometre radii

Simplified route representation – only "from / to"

Departure days, hand-overs and restrictions are barely taken into account

No transparent view of local over- or underperformance



40+
countries in the
network

3
modes of transport:
road, air, sea

THE CENTRAL CHALLENGE

How can we model real, up-to-date traffic so that lead times become **realistic, transparent and reliable**?

Sales

Dispatch

Customer Service

Our customers

Rethinking the backbone of transport planning

Build a prototype for managing and using **line-haul schedules** between DACHSER branches – the network not as a static plan, but as a living traffic network.

01

Traffic relations

Model "from – via – to"
instead of just origin and
destination

02

Node types

Take platforms, hubs and
hand-over locations into
account

03

Control lanes

Activate or deactivate
traffic lanes

04

Validities

Work with periods – valid
from / valid to

05

Lead-time calculation

Plausible lead times from
A to B – transparently
composed

Strategic context: **Interlocking Recharged / Global Groupage** – chaining line hauls across different platforms and business fields.

Shipments change business field en route



Pallets are handed over to a **ground handler** at the airport or a **shed operator** at the port. Additional time arises there:

- Customs clearance

- Transshipment

- Consolidation / palletising (CFS – container freight station)

Your prototype should make these hand-over points visible and account for their effect on total lead time.



3 pallets, Istanbul → Kempten

What does the lead time actually consist of? Every block is a node or an edge in your model.



The problem is not the deviation – it is that today nobody can explain it.

WHAT YOUR PROTOTYPE SHOULD DO

Five building blocks – covered by example

1 Routes & distances

Real distances and driving routes
GPS waypoints based on real data
Average speeds from historical data
POIs: swap-body exchange and exchange times

2 Restrictions & calendars

Weekend and public-holiday logic per country
Regional driving bans: federal states, cantons, districts
Manage country-specific driving times

3 Departures & schedules

Fixed departure days and times
Traffic frequencies like a bus timetable, valid per weekday
ASL sailing schedules from carriers and airlines

4 Products & lead-time logic

entargo, vivengo and avigo calculate on the same basis
Take cut-off times into account
Manage maximum lead-time tolerances
Priorities: speed, flex, fix

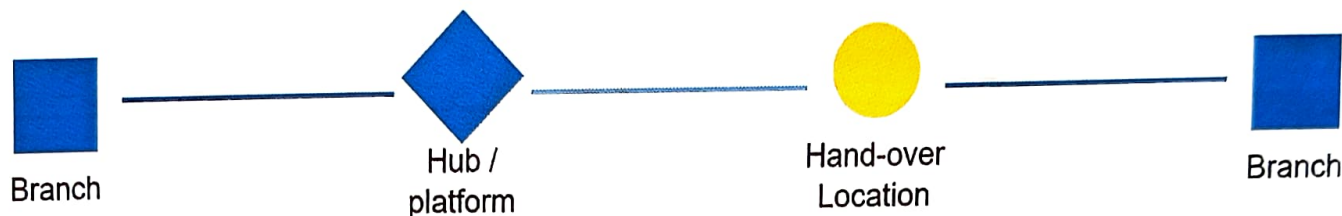
5 Visualisation

Map view A – B – via
Simulate a timeline / lead-time bar
Status active / inactive
Options: how the service can be produced

Important

This is not about completeness or perfect data, but about a plausible, well-considered model that covers all requirements and relates them to one another.

The schedule as a graph



Nodes

Branches · platforms · hubs · hand-over locations with typical dwell time

Edges

Traffic lanes for road, air and sea – each dependent on the next connecting service

Every lane carries

Transport mode

Departure days / times

Lead time

Arrival times

Restrictions

Status active / inactive

Validity valid from / to

You decide – and you may simplify

You decide yourselves

How deep you go technically

How you model data – graph, services, UI, simulation

How you visualise and present the result

The focus is on the data model, logic and visualisation – not on perfect data.

Explicitly allowed

Mock or sample routes

Limiting yourselves to three or four countries

A few example rules, e.g. the German Sunday driving ban

Hand-over location as its own node type with a static dwell time

What does matter is the dependency on the next connecting service.

Not part of the task

Completeness of the network

Perfect or production-grade data quality

Integration with existing DACHSER systems

Pricing or rating

Data sources as inspiration: historical shipment routing, relation mapping tables, departed / arrived events, daily volumes.

So everyone means the same thing

Line haul

Scheduled long-distance service between two DACHSER sites, e.g. overnight Kempten → Hub Überherrn.

Interlocking

Interlinking Road Logistics and Air & Sea into one chain. "Recharged" = the new edition.

Platform / hub

Transshipment point where shipments from many line hauls are redistributed.

Sailing Schedule

Fixed sailing or flight schedule from carriers and airlines – an external given.

Mock route

An invented but realistically calculated sample route – allowed and wanted.

Local transport

Pickup and delivery in the area around a branch – the first and last kilometre.

Hand-over Location

Where the business field changes: hand-over to a ground handler or shed operator.

Lane

A connection between two nodes – the edge in the graph model.

POI & swap time

Point where swap bodies are exchanged; the swap time is the standing time required.

Graph model

A network of nodes (sites) and edges (lanes) – enables route comparison.

Groupage / LTL

Shipments from several customers travel bundled: 3 pallets instead of a full load.

CFS

Container freight station: the container is unloaded and the goods re-palletised (consolidation).

Cut-off time

Latest acceptance time for a departure: miss 17:00 and you lose a day.

entargo / vivengo / avigo

Product families for European groupage, sea freight and air freight.

Driving ban

Statutory closed period for trucks, e.g. Sundays in Germany – costs a full 24 h.

THE TARGET PICTURE

What stands at the end of the two days

Your prototype shows

How real transports actually flow through the network

Which route is chosen – and what options exist

How lead times are composed, step by step

How that becomes a reliable customer promise

Deliverables

A working prototype – data model, logic, visualisation

A pitch to the jury, clearly explainable

Short documentation – model assumptions and simplifications

Using AI to build is explicitly encouraged – the result has to be feasible in practice.

How we assess

Robustness of the data model

Logical consistency of the lead-time calculation

Plausibility rather than completeness

Comprehensibility of the visualisation

Explainability in the pitch

Feasibility in practice

This creates the basis for better transport planning, real-time shipment visibility and calculation foundations for digital customer products such as the ProductPilot.

"If we want real-time logistics, we must first understand real networks."